

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — FULL STACK ACTIVE (OPEN ECOSYSTEM)

Ecosystem, Open Toolchains & Future Frontiers

1. Open Toolchain

- OpenROAD Physical EDA
- Verilator Fast Simulator
- SkyWater 130 nm / OpenPDK
- RISC-V Open ISA Standard

2. Community Hub

- Shared Model Weight Hub
- Open Benchmark Suite
- Standard Data Schemas
- Collaborative Provenance

3. Future Frontiers

- Autonomous Chip Tapeout
- Wafer-Scale AI Systems
- Photonic & Quantum Co-Design
- Continuous In-Field DSE

4. Moonshot Target

- Verified Architecture Result
- Sustainable Semiconductor Future

Community Feedback & Open Ecosystem Refinement Loop